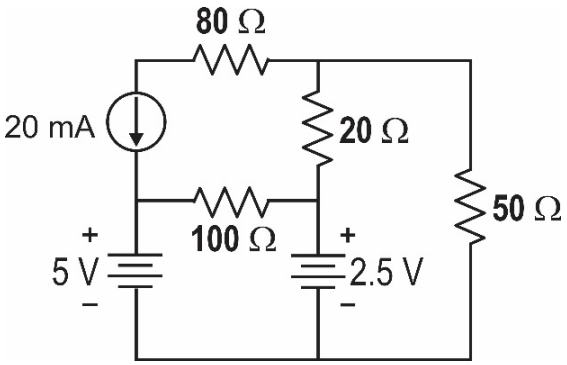
NAME _____ NetID _____

ECE-UY 2004 **MIDTERM EXAM #2** – 3 Problems, 1 Extra Credit **SAMPLE Fall 2026**

READ THESE INSTRUCTIONS

- **RETURN this sheet with your blue book**, **FIVE** points deducted is you fail to return exam sheet.
- Enter your name and NetID. (not N#)
- This is a closed book exam, no notes or paper of any kind – **Reference Sheet Attached.**
- Scrap paper will be provided upon request.
- **SHOW ALL YOUR WORK, NO POINTS FOR JUST AN ANSWER ON ANY STEP**

1) Nodal Analysis (10 pts): Use nodal analysis to answer the following questions. All sources are independent.



A) **Assign voltages to each node** in the circuit (include the ground reference). You can redraw the circuit in your exam book or write the node voltages on this exam paper. (1 pt)

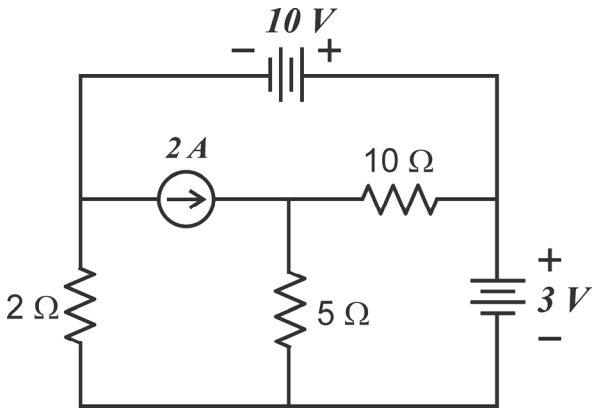
B) What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$). You will be using this equation in (C).

C) Write a sufficient **set of equations** to solve for all the node voltages. Clearly list your complete set of equations. (6 pts)

D) Solve the equations in (C) for the node voltages. Provide a **Table** of all the node voltage values. (1 pts)

E) Calculate the **power** in the current source. Determine whether the source is delivering or absorbing power. (2 pts)

2) Mesh Analysis using Super-Mesh (10 pts)



A) **Assign mesh currents.** You can redraw the circuit in your exam book or draw the mesh currents on this exam paper. **Clearly label all the mesh currents and identify the Super-Mesh with its direction (CW or CCW)**. (1 pt)

B) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

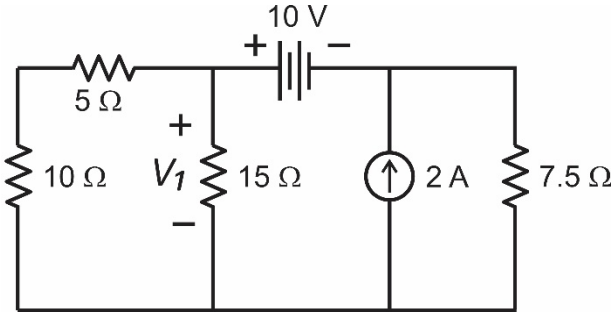
C) Write a sufficient **set of equations** to solve all the mesh currents. Clearly list your complete set of equations. (6 pts)

D) Using the equations in (C), solve for the unknown **mesh currents.** Provide a **Table** of values for <u>all</u> mesh currents. (1 pts)

E) Calculate the **power** in all three sources. Determine whether the source is delivering or absorbing power. Place your answers on a table. (2 pts)

Next page for problems 3-4

3) Superposition (10 pts)



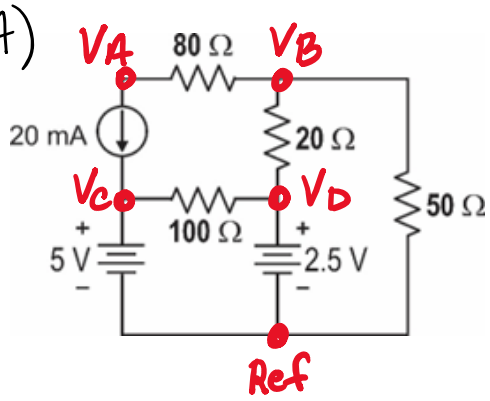
Using Superposition, determine the voltage V_1. Clearly show all steps in your solution.

4) EXTRA CREDIT Not Shown

END OF MIDTERM EXAM 2

Nodal

A)



B) $\sum I_{in} = 0$

C) we know ; $V_C = 5V$, $V_D = 2.5V$
only two unknowns, V_A, V_B

KCL at A

$$-20mA + \frac{V_B - V_A}{80\Omega} = 0$$

multiply by 80Ω

$$-1.6V + V_B - V_A = 0$$

$$-V_A + V_B = 1.6V \quad (1)$$

KCL at B

$$\frac{V_A - V_B}{80\Omega} + \frac{V_D - V_B}{20\Omega} + \frac{0 - V_B}{50\Omega}$$

Multiply by 400Ω

$$5V_A - 5V_B + 20V_D - 20V_B - 8V_B = 0$$

$$5V_A - 33V_B + 20V_D = 0$$

$$5V_A - 33V_B + 20(2.5V) = 0$$

$$5V_A - 33V_B = -50V \quad (2)$$

Equations

$$V_C = 5V \quad (1)$$

$$V_D = 2.5V \quad (2)$$

$$-V_A + V_B = 1.6V \quad (3)$$

$$5V_A - 33V_B = -50V \quad (4)$$

D) Solve

multiply (1) by 5 and add to (2)

$$-5V_A + 5V_B = 8V$$
$$5V_A - 33V_B = -50V$$
$$0 \quad -28V_B = -42V$$

$V_B = 1.5V$ (3)

plug (3) into (1)

$$-5V_A + 5(1.5V) = 8V$$

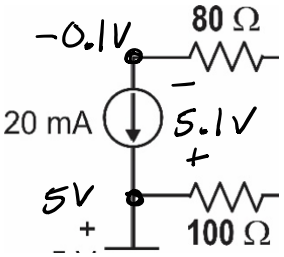
$V_A = -0.1V$ (4)

Table :

V_A	$-0.1V$
V_B	$1.5V$
V_C	$5V$
V_D	$2.5V$

V_C	5 V
V_D	2.5V

D)

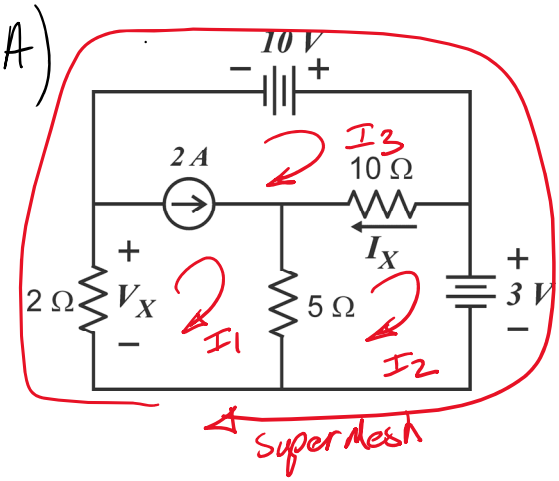


$P_{cs} = (20mA)(5.1V) = 102\,mW$ delivered

current leaving "+"

SuperMesh

A)



B) DROPS

C) Supermesh KVL Equation

$2\Omega I_1 - 10V + 3V = 0$

$I_1 = 7V/2\Omega = 3.5A$ (1)

Constraint equation

$2A = I_1 - I_3$ (2)

Loop 2 KVL

$5\Omega(I_2) - 5\Omega(I_1) + 10\Omega(I_2) - 10\Omega(I_3) + 3V = 0$

$-5\Omega(I_1) + 15\Omega(I_2) - 10\Omega(I_3) = -3V$ (3)

D) plug (1) into (2)

$I_3 = 1.5A$ (4)

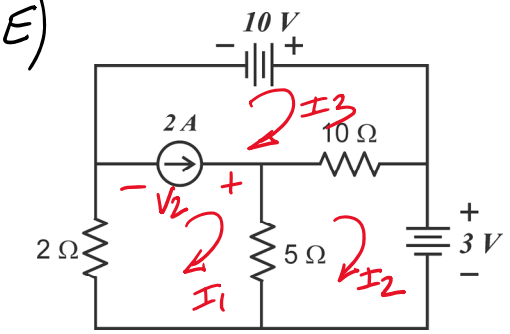
plug (1) and (4) into (3)

$I_2 = 1.967A$ (5)

Table

I_1	3.5A
I_2	1.967A
I_3	1.5A

E)



$P_{3V} = (3V)(I_2) = 5.9W$ Absorbed

$P_{10V} = (10V)(I_3) = 15W$ Delivered

Solve for V_2 using KVL around Lower left

$2\Omega(I_1) - V_2 + 5\Omega(I_1) - 5\Omega(I_2) = 0$

$V_2 = 7\Omega(I_1) - 5\Omega(I_2)$

$= 7\Omega(3.5A) - 5\Omega(1.967A)$

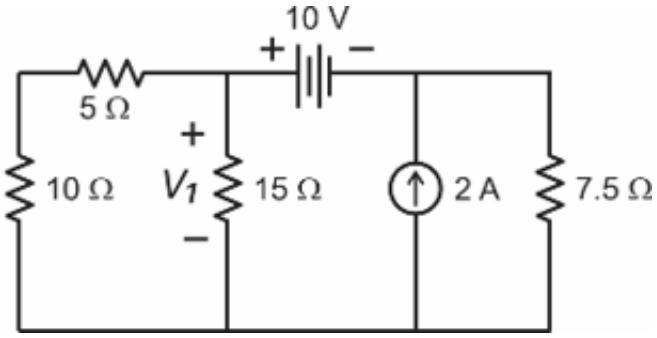
$= 24.5V - 9.833V$

$= 14.67V$

$P_{2A} = (V_2)(2A) = (14.67V)(2A)$

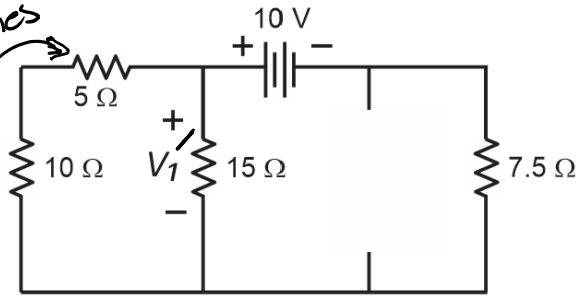
$= 29.33W$ Delivered

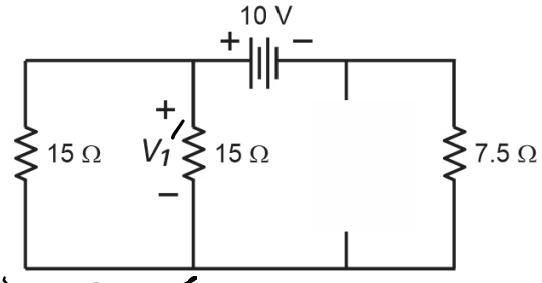
Superposition



step ① Keep 10V, I→0 (open)

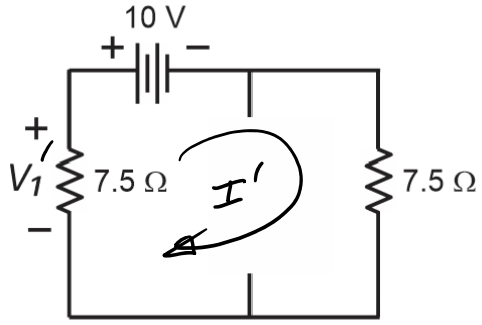
series





parallel

same V_1'



KVL CW

$7.5\Omega I' + 10V + 7.5\Omega I' = 0$

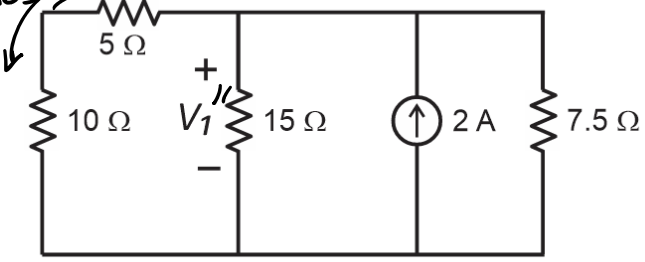
$I' = \frac{10V}{15\Omega} = 666.67mA$

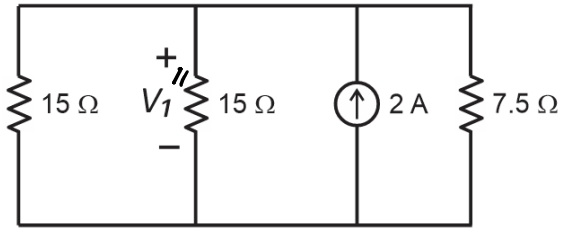
$V_1' = 7.5\Omega(I')$

$V_1' = 5V$ (1)

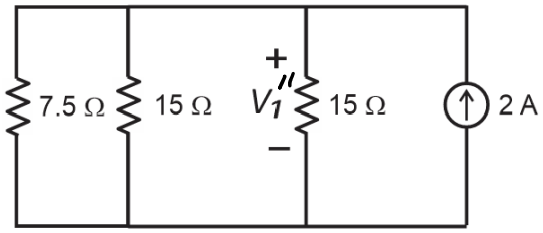
Step ② Keep 2A, V→0 (short)

series

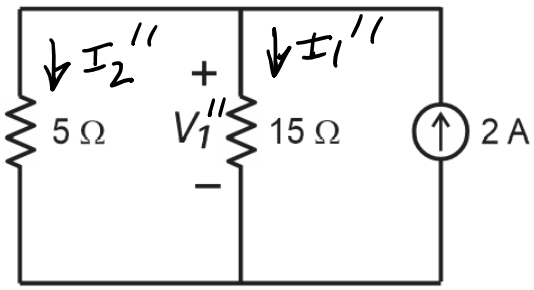




All parallel rearrange



parallel combine



use current divider

$$I_1'' = 2A \frac{5\Omega}{15\Omega + 5\Omega} = 500mA$$

$$V_1'' = 15\Omega\,(I_1'') = 7.5V$$

Step (3)

$V_1 = V_1' + V_1''$

$\boxed{V_1 = 12.5V}$

NOTE: There are many ways to solve each step including NODAL and Mesh